

TECHNICAL SPECIFICATION FOR 30 V. 35A MANUAL AND AUTOMATIC FLOAT-CUM-BOOST BATTERY CHARGER

1. Scope: The specification covers the design, manufacture, testing at the Manufacturer's works, delivery by road transport to different sites of West Bengal State Electricity Distribution Company Limited for Manual and Automatic Float cum Boost Battery Charger. 30 V DC output in Float Mode: Continuous 21 Amp load and trickle charging current for Battery. Boost Charge Mode: Continuous load of 21 Amp and Battery Boost Charge current of 14 Amp. In addition to above, the Charger is capable of delivering short time current of 40 Amp for 1 second for tripping of breaker and lighting purpose.

Supervision and erection of devices shall have to be undertaken by manual acceptance of terms and conditions for the same, if required.

APPLICATION:

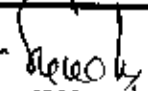
The system requires a reliable and uninterrupted D.C. supply for supplying D.C. Power to emergency lights, closing and tripping coils of circuit breakers, relays etc.

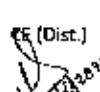
2. Service conditions:

Equipment to be supplied against this specification shall be suitable for satisfactory continuous operation under the following tropical conditions.

Maximum ambient temperature (Degree C)	50
Maximum temperature in shade (Degree C)	45
Minimum Temperature (Degree C)	3
Relative Humidity (percent)	95
Maximum Annual rain fall (mm)	1450
Maximum wind pressure (kg/sq. m)	150
Maximum altitude above mean sea level(Meter)	1500
Isoceraunic level (days per year)	50
Seismic Level	The sites fall within seismic zone-III and IV as classified in the IS:1983
Moderately hot and humid tropical climate conducive to rust and fungus growth	

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3.0 Standards:

Unless otherwise specified, the equipment shall conform to latest applicable Indian standard of equivalent IEC, British or USA standard and in particular to the following standard (or equivalent IEC British, USA standard):

IS:3895/1966	Specification for the rectifier equipment
IS:9224	Specification for HRC fuses.
IS: 1248	Indicating instrument.
IS:375 /1963	Specification for wiring
IS: 4540/1968	Mono crystalline semiconductor rectifier assemblies
IS:2026	Transformers
IS:13947/ 1993	Air Break Switch/Contactor
IS: 5/1978	Colour for ready mix paint
IS: 5421/1981	Printed circuit board
IS: 8828/1993	Miniature circuit breaker
IS:2147	Degree of protection for cubicle
IS:6619	Safety code for semiconductor rectifier equipment

The tenderer shall clearly state the standard to which the equipment offered by him conforms.

4.1 SUPPLY: AC input: 240 volts +20 % & - 25%, 50 HZ +/- 5%, Single phase supply. Input power factor 0.7 minimum at rated load.

4.2 CHARGER SYSTEM OPERATION:

4.2.1. The Battery Charging Equipment shall be float-cum-Boost type with facility to supply the DC continuous load of 21 Amp also. During normal operation, the Battery is floated across the Battery charger at 33.75 V (2.25 V per cell) and should be compatible for battery as per specification and also supplies the load current 21 Amps. Whenever the battery has run down, it needs to be boost charged passing higher current of 14-Amps; into batteries for higher voltage up to 42-V (2.75V maximum per cell). The charger shall provide this extra voltage for Boost charging.

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4.2.3 Battery Charging Equipment shall be fully automatic for Float and Boost charging facility with suitable indication on front of the panel by means of LED indication. For this, automatic solid state changeover relay shall be provided. During AC mains fail, Battery supplies the load and gets discharged. The extent of discharge depends on the duration of power failure and the current drawn by load. When the mains are restored after power failure, Battery shall put on automatically on boost mode. During this time, load voltage shall be maintained by load regulator $30\text{ V} \pm 2\text{ Volt}$. After the Battery gets fully charged i.e. its voltage reaches 42 Volts, it should automatically switch over to trickle mode & when Battery voltage falls below 27 volts, the charger should automatically switch over to Boost mode. Soft start feature shall be provided in the system.

4.2.5 The battery charger shall have one top and one bottom for Manual & Auto Bridge circuit. The bottom module (Auto mode) shall be so connected to the top/other modules that if necessary, it can be taken out for repair by simply disconnecting few links. Each arm of the bridge shall be provided with filter circuit as well as fuse protection in order to rectify the fault if any, at the earliest opportunity.

4.2.7 Automatic changeover of DC load requirement of substation to Battery in the event of Mains power failure should be without any break. I.e should be made without incorporating any relay in the output DC supply.








4.2.8 Automatic changeover of output to charger supply with restoration of incoming power supply and after meeting short time current requirement from battery.

4.3 **RATING:** 30V DC output, 35 Amp Auto & Manual dual automatic float cum boost Battery Charger.

a) Type : Thyristor controlled

b) At Load Terminal:

During Float charging or Boost charging, the load voltage shall be maintained $30\text{ V} \pm 2\text{ Volt}$ for AC input voltage variation of +20% & -25 % and load variation of 0 to 100 %. Also system shall provide 35 Amps continuous current and also momentary current of 40 Amps for one second.

c) At Battery Terminal:

Trickle charge voltage 33.75 volts (2.25V per cell) at 400 mA maximum.

Boost charge voltage 42 volts (2.75 V per cell) at 14 amps.

d) RIPPLE:

The ripple content in the DC output of Battery Charger shall be limited to 2 % RMS.

e) Regulation: +/- 2%

f) Efficiency: More than 70%

g) Method of cooling : Natural air with proper ventilating arrangement

5.1 COMPONENTS:

5.1.1 The Battery charger shall comprise of following components but not limited to the same:

a) Double pole AC circuit breaker for AC incoming of the battery charger (MCB) with auxiliary potential free contact for indication to SCADA.

~~b) Single phase, double wound, Air Natural Cooled, Vacuum impregnated transformer of CRGO lamination and copper winding.~~

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not be any discontinuity of DC supply to the bus bar during any transition period and battery power should be available for tripping circuit, if necessary, even during boost charging.

- o) AC analog Voltmeter 96X 96 sq.mm of rating 0 to 300 V.
- p) MCB Trip indication lamp/ LED for load and Battery MCB.
- q) The indicating instruments shall be of class 1.0 accuracy.
- r) The following provisions conforming to relevant IS shall be made on the front panel:-
 - (i) Voltmeter to indicate battery/ charger DC voltage.
 - (ii) Voltmeter for Input AC supply voltage.
 - (iii) Ammeter to indicate Charge/Discharge current of Battery.
 - (iv) Ammeter to indicate load DC current.

5.1.2 The Charger shall be provided with following LED of reputed make
Indication:

- (i) Supply of power -- Green
- (ii) Charger on -- Green
- (iii) Battery reverse polarity
- (iv) Input power supply fail – Red
- (v) Output over/under voltage
- (vi) Earth fault

5.1.3 Audio/ Visual alarm to indicate:-

- (i) Power failure.
- (ii) Charger failure.
- (iii) Battery disconnection/failure.
- (iv) DC under/Over voltage.
- (v) Condenser Fuse failure.
- (vi) In case of failure of charger on fault, it should give buzzer as well as LED indication. However, the buzzer alarm should be provided for battery MCB fail, load MCB fail and charger trip due to overload indication with an accept and reset switch. The charger shall have provision for an alarm relay contact for remote Indication. Suitable terminals with Identification label shall have to be provided.

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5.1.4 Controlling arrangement for following functions shall be provided:-

- a) Auto/Manual Selector Switch
- b) Manual operation controlling device
- c) Accept/Reset push button
- d) Voltmeter selector switch

5.2 a. Wiring: Charger cubicle shall be supplied completely wired upto terminal block for purchaser's external connection using solder less crimping type copper lugs. All wiring shall be carried out with 750 V grades PVC Insulated multi-strand copper conductor of 2.5 Sq mm and shall be flame/ vermin proof. All wiring shall be neatly bunched without affecting access to equipment/ components mounted within the cabinet. The Charger DC output circuit for Battery and load connection separately shall be wired with 6.00 sq.mm PVC insulated cable.

Suitable two earthing terminal and outgoing terminal shall be supplied to connect the external supply cables.

Control/Indication/Annunciation circuit shall be wired with suitable size of PVC insulated cable as per scheme requirement. Colour coded wires should be used to facilitate easy tracing, as under:-

i. Single Phase A.C. Circuit:-

- (a) Yellow for Phase
- (b) Green for Earthing
- (c) Blue for Neutral

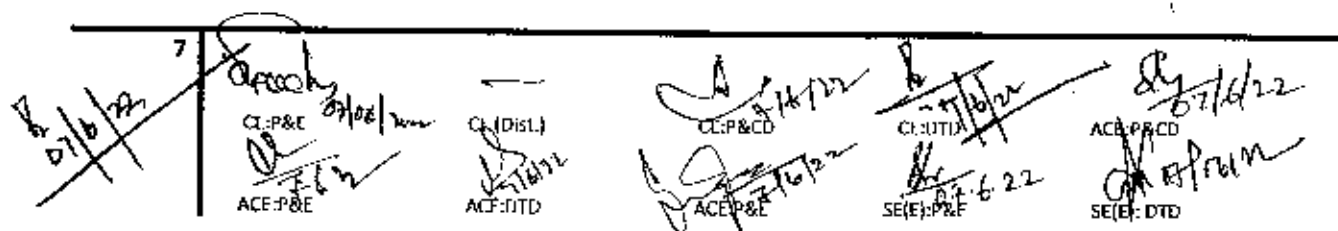
ii. D.C. Circuit:-

- (a) Red for Positive
- (b) Black for Negative

iii. Control Wiring:-

- (a) Gray for annunciation and other control circuits.

b. Ferrules: Embossed/Engraved core identification ferrules, marked to correspond with the wiring diagram shall be fitted at both ends of each wire.



c. Termination:

- The input, battery and load terminals shall be located in easy accessible positions. The terminals shall be properly labelled for easy identification of Input (Ph & Neutral), Battery (positive & negative), Load (Positive & negative) and Emergency Lamp (Positive & negative).
- The terminals shall be rated at 25 Amps or more depending on scheme requirement.
- The Input terminals shall be connected to a 3 core PVC insulated multi-stranded copper conductor cable (minimum 2 mtr. length) with a 25 Amp 3 pin plug socket at the other end. The earth conductor of the cable shall be provided on the charger body.
- The charger shall incorporate terminals and fuse of 4 Amp rating for connecting wires from outside to bring emergency light into battery circuit in the event of mains failure.
- All control cables should run through the bottom side of the Charger cubicle with proper gland arrangement.

5.3 Special Feature:

a) All printed circuits boards used in the Battery charger shall be solder marked, glass, epoxy, FR 4 grade copper clad material having edge type gold plugging connectors conforming to latest IS-IEC specifications. All assembled PCB conforming coating on component side & epoxy varnish on the other side with suitable protective coating for protection against humidity and corrosion.

b) Transformer shall be with class-F insulation having a continuous rating at least 125% of the rating of the charger. Reactance of the transformer shall be suitable to take care of regulation and surges. The power transformer unit of the battery charger shall be designed for adequate VA rating but in any case it should not be less than 3000 VA and should be rated for 300 V with adequate factor of safety. The heat dissipation and power control system should be designed so that junction temperature should not increase 60°C over ambient 50°C. Rating of silicon diode should not be less than 63 A (RMS).

Please note that necessary documentary evidence, showing transformer rating of 3000 VA along with test certificate from manufacturer, if bought-out, shall be enclosed, for approval of the purchaser.

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c) Tenderer shall furnish full description and illustration of all components complete with quantity, make, rating, type, tolerance etc. including, but not limited to the list of components given below.

d) Following make of components shall be used in the Battery charging equipment approved by C.E. (P&E) WBSSEDCL.

i)	Switches	Kaycee/ Recom/L&T
ii)	Meters	AE/IMP /RISHAV/UNIQUE
iii)	Contacts	L&T/Siemens/ABB/C&S
iv)	MCB	MDS/ABB/Siemens/L&T
v)	HRC fuses	S&S/Bussman/Siemens/L&T/HBL
vi)	SCR	HIRECT/IR/Semicron/RIR
vii)	Diode	IR/Semicron/RIR/HIRECT
viii)	Lamps/LED	Essen/vaishna/Siemens/L&T/Emco
ix)	Filter condenser	Rescon/Alcon/Kendil/Keltron
x)	Potentiometer	Pankaj/Boorns
xi)	Voltmeter A.C.	IMP/AE/Rishav/ Meco
xii)	Relays	PLA/OEN/OMRAN/GOOD SKY

e) All other bought out items shall conform to the relevant IS and of reputed make subject to approval of C.E. (P&E) WBSSEDCL.

f) a) The charger cubicle will be indoor type with all associated and auxilliary equipments mounted therein.

b) The charger cabinet shall be 1.6 mm. thick sheet steel for Non – Load bearing members & 3 mm for Load bearing members enclosed and shall be dust and vermin proof. The maximum overall dimension of the charger shall be height-1800 mm(including the height of Stand), Width – 700 mm and depth - 600 mm.

c) The cabinet shall be free standing floor mounting type with a provision for proper ventilation. Backside Cover will be double door type. Necessary arrangement on front, back & top side shall be provided for easy access to different components for repair/maintenance etc. The enclosures of the cabinet shall provide at least degree of protection IP-42 as specified in IS: 13947.

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- d) All doors, removable covers and plates shall be provided with neoprene gasket all around. Louvers, when provided shall have screens and filters. The screen shall be of fine wire mesh made of G.I. wire.
- e) All sheet steel work shall be degreased, pickled phosphate and then applied with two coats of Zinc primer and two coats of finishing epoxy powder coating or polyethylene protective coating of light gray as per IS : 631, both inside and outside. The cabinet legs shall be of adequate height and strength and should provide minimum clearance of 300 mm from ground.
- f) All fuses shall be of HRC cartridge types conforming to IS: 2208 mounted on plug type fuse bases.
- g) Electrical indicating instruments shall be flush mounted on panel with only flanges projecting. Dial shall be white with black number and lettering.
- h) Control and instrument switches shall be of rotary type.
- i) Indicating lamps shall be LED type with low watt consumption. The LEDs shall be provided with series resistors.
- j) Strip type space heaters of adequate capacity shall be provided inside cabinet to prevent moisture condensation.
- k) All door mounted equipments as well as equipment mounted inside the cabinet shall be provided with individual labels with equipment designation engraved on aluminium/plastic plate (stickers are not acceptable).

l) **POLARITY MARKING:-**

The polarity marking of the terminals shall be marked for identification. The positive terminal may be identified by "P" or (+) sign or **Red** colour mark and the negative terminal may be identified by "N" or (-) or **Blue** colour. Terminal marking shall be permanent and non-deteriorating.

m) **TERMINALS**

Battery charger cabinet shall be provided with two separate suitable earthing terminals. Separate terminals shall be provided for connecting load and battery leads

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The following documents shall be necessarily enclosed along with Tender Submission:

- n) All Deviations from this specification shall be set out by the bidder and should be brought out clearly in the Deviation sheet.

6.0 Technical Particulars of Matching Lead Acid Cell Battery for 30 V, 100 AH

15 nos. 2 volts, 100AH at 27 Deg C when discharged at 10 hours with final cell voltage not less than 1.85 Volts.

- a) Range of Float Charging : 80mA to 240mA
- b) Range of Boost charging
 - i) Starting Current : 14 Amp
 - ii) Finished rate : 7 Amp.
- c) **Cell voltage at the end of charge**
 - i) Float charge condition : 2.25 V per cell
 - ii) Boost Charge condition : 2.75V per cell
- d) **Permissible cell voltage at the end of discharge** : 1.85V

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6.2 Ampere Hour capability of each cell of Battery

Amp-Hour Capacity	100AH
Discharge duration	10H
Final Voltage	1.85V

7.0 Nameplate Information

Battery Charger nameplate information shall be determined in agreement with the Employer. This information may include for example:

- Name of manufacturer and country
- Type, design, and serial number
- Rated frequency
- Rated Output Voltage and Current
- Purchase Order number and date
- Month and year of supply
- Property label: 'Property of WBSEDCL'
- Guarantee : Five years

8.0 Type Tests:-

The following tests shall constitute the type tests as per IS: 3136-1965 and IS:4540-1968 (latest amended):

- a) DC voltage current test.
- b) Automatic voltage regulator operation.
- c) Efficiency test.
- d) Ripple Voltage test.
- e) High Voltage test.
- f) Temperature rise test.

9.0 List of Routine Tests as follows:-

- Physical checking of charger as per approved drawing.
- ~~Insulation resistance test.~~
- High voltage test (2.0 KV).
- Line regulation test at No Load and full load in test at load in Float/ boost mode.

12

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- Full load ripple content measurement test at load terminals in float and boost mode.
- Voltage regulation test at load terminal and Battery terminals in float and boost mode.
- DC short circuit test without blowing HRC fuses and without tripping MCBs at load terminal and also at Battery charger output terminals with observation of total current limit.
- Efficiency measurement test at full load and 20% load.
- Checking of automatic operation Float to Boost, Boost to float mode as per Battery condition.
- Checking of Battery voltage, load voltage and Load current Boost Mode at different charging current.
- Temperature rise test of complete charger at full load current (21+14) Amp @ ambient temperature.
- Checking of battery & Load Terminal voltage in manual Mode operation at different position of Manual Mode Selector Switch.
- Checking of automatic connection of Battery to load in case of 'Mains Failure' or Charger Trip' conditions.
- Checking of Float voltage setting range and Boost current setting Range in Automatic Mode.
- Any other routine test shall be carried as per IS: 4540 on the complete battery Charging equipment.

9.1 For acceptance of lot, all above routine tests shall be carried out.

10.0 Testing Facilities:

The Bidder must clearly indicate what testing facilities are available in the works of manufacture and whether the facilities are adequate to carry out all Routine, Acceptance Tests. These facilities should be available to WBSEDCL's Engineers, if deputed to carry out or witness the tests in the manufacture's works. If any of the tests can not be carried out in the manufacturer works, Bidder shall have to arrange

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for such testing at any NABL Accredited/ Govt. Test House or Laboratory at Bidder's cost.

11.0 **Inspection :**

All tests and inspection shall be made at the place of manufacturer unless otherwise specially agreed upon by the manufacturer and WBSEDCL. The manufacturer shall provide the WBSEDCL all reasonable facilities, without charge to satisfy him that the material is being supplied in accordance with this specification.

12.0 **Tests:-**

General: The equipment including all components and accessories shall be subjected to all type of tests Including Routine and acceptance tests in accordance with provision contained in relevant standard.

12.1 Type Test: The Bidder shall have to submit along with their Tender documents, as pre-requisites, the complete Type test Reports as stipulated as per Clause No.8.0 of the specification, carried out within 5 years from the due date of Tender, from CPRI/NABL accredited/Govt. recognised Test House or Laboratory on the offered Item, failing which their offer may not be technically acceptable. However, in case of failure to submit the same, submission of type test report on 20.5 Amp Manual & Automatic Float-cum-Boost Battery Charger during submission of Tender is essential.

If the Tenderer fails to produce the said type test report on the offered item during submission of tender, he will have to submit the said type test report before opening of price bid at his own cost.

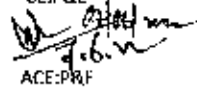
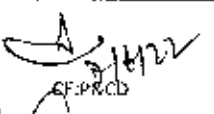
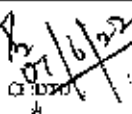
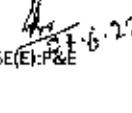
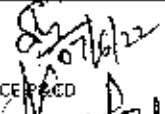
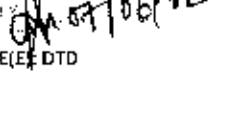
12.2 Routine and acceptance Tests: Routine & acceptance tests shall have to be carried out in compliance with provision contained in the relevant standard and / or to ascertain satisfactory performance of the offered device at the works of the Manufacturer.

The acceptance tests shall have to be conducted in the presence of authorised representative of the WBSEDCL before effecting delivery.

13.0 **Drawings:**

The successful bidder shall have to submit the 6(six) copies of drawings, manuals, literatures for approval to the office of the Chief Engineer (P & E) for approval

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before starting manufacture of the equipment. It is desired that the complete schematic of the charger is provided on a permanently laminated/engraved plate of suitable thickness which has to be bolted/riveted at the four corners on the inside face of rear door. In addition, one more plate of similar type and dimension shall be provided on the outside of the rear door providing guidelines and instructions for operation of the charger. The guidelines and schematic to be provided on the plates shall be as per approved drawings.

Before despatch of the equipment, 6(six) copies of drawings, manual and literature shall be submitted to WBSEDCL for distribution to different offices of the Company. In addition to above, every crate of complete set of equipment shall also contain in waterproof folder, 1(one) set of drawing, manuals and literature for commissioning, operation and maintenance at site.

14.0 PERFORMANCE GUARANTEE:

Battery Chargers shall be delivered to the various consignees of WBSEDCL and shall be suitably packed to avoid damages during transit. The Battery Charger with all its integral part will be guaranteed for the period of 5 years from the date of last dispatch.

In the event of any defect in any integral part of the equipment arising out of faulty design, materials, and workmanship within the above period, the supplier shall guarantee to replace or repair the same to the satisfaction of the WBSEDCL.

However, any engineering error, omission, wrong provision, etc. which do not have any effect on the time period, shall be attended to as and when observed/pointed out without any price implication.

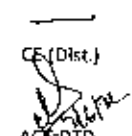
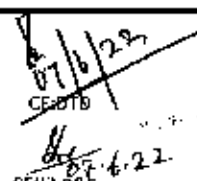
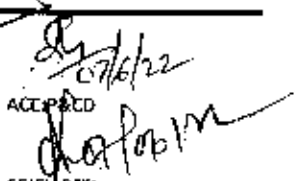
15.0 GUARANTEED TECHNICAL PARTICULARS:

The bidder should fill up the details in schedule A - 'Guaranteed Technical Particulars' and the statement such as "as per drawing enclosed", "as per WBSEDCL requirement", "as per IS", "as per specification" etc. shall be considered as details not furnished and such offers will be rejected.

16.0 Schedules:

The Bidder shall fill in the following schedules, which is part and parcel of the tender specification and offer. If the schedules are not submitted duly filled in with the offer, the offer shall be liable for rejection.

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17.0 QUALITY ASSURANCE PLAN:

17.1. Immediately on receipt of this order you shall have to submit a "Quality Assurance Plan" indicating the specific quality control procedure and practices adopted in the major activities of production to ensure its standard.

17.2 The successful Bidder shall within 30 days of placement of order, submit following information to the purchasers.

(i) List of raw material as well as bought out accessories and the names of sub suppliers selected from those furnished alongwith offer.

(ii) Type test certificates of the raw material and bought out accessories.

(iii) Quality Assurance Plan (QAP) with hold points for purchaser's inspection. The quality assurance plan and purchaser's hold points shall be discussed between the purchaser and supplier, before the QAP is finalized.

17.3 The successful Bidder shall submit the routine test certificates of bought out accessories and central excise passes for raw material at the time of routine testing of the fully assembled equipment.

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Schedule 'A'

GUARANTEED TECHNICAL PARTICULARS OF 30V, 35 A (14A+21A) BATTERY CHARGER SUITABLE FOR 30 VOLT 100 AH LEAD ACID PLANTE TYPE BATTERY

Sl. No.	Description	Particulars
1	Name of manufacturer	
2	Type of charger	
3	Indicate specification to which the offered charger conforms	
4	Rated Input supply voltage	
5	Rated Input current	
	Input Power Factor	
6	Output voltage with range of variation	
7	Output current range	
8	Output current a) Max. Battery charging current b) Max. continuous DC external load current c) Max. continuous output current	
9	Output voltage Regulation a) No load b) Full load	
10	Ripple factor	
11	Efficiency	
12	Method of Cooling	
13	Painting details Paint shed Paint thickness	
14	Material of enclosure	
15	Degree of Protection of enclosure	
16	Whether Backside Cover will be double door type(Yes/No)	
17	Whether separate terminals of M-12 size for connecting load and charger leads to the battery terminals has been provided.	
18	Provision of LEDs for indication purpose	
19	Provision of Audio/ Visual alarm	

17

CE:P&E

ACE:P&E

CE:DTD

ACE:DTD

CE:P&CD

ACE:P&E

CE:DTD

SE(E):P&E

ACE:P&CD

SE(L):DTD

20	Provision of labels for individual components	
21	Provision of ferrules	
22	Provision of separate terminals for connecting battery and load	
23	Provision of two separate Earthing terminals	
24	Weight of the complete charger	
25	Overall dimensions of Battery Charger(HxWxD)	
26	Thickness of the sheet enclosure (Load Bearing Member and Non-Load Bearing Member)	
27	Make and rating of Miniature Circuit Breaker for AC Input supply	
28	Make & rating of main transformer	
29	Make and type of rectifier SCR	
30	Rating of rectifier SCR in Amps	
31	PIV value of rectifier diode	
32	Output in Amps of complete rectifier stack	
33	Make, Type & range of Voltmeter	
34	Expected service life of Charger	
35	Commitment for availability of spares during service life	
36	Applicable Standards	
37	List of accessories to be provided with the Charger	
38	Recommended particulars of matching Battery	
39	Technical particulars of different components including make, type and rating	
40	Description of operating principle of Auto/Manual float cum boost charger with functions of different components supported by drawing and block diagram	
41	Guarantee	

Place:

Signature with Date:

Name of the Company: _____

Name & Designation: _____

Seal of the Company:

18

CE:P&E
ACE:P&E

CE(Dist.)
ACE:DTD

CE:P&E
ACE:P&E

CE:DTD
SE(E):P&E

AC:PRCD
SE(E):DTD